An Exact All-Support Matrix Algorithm for the Optimal One-Level Test Function

Candidate full solution to the coefficient problem in arXiv:1507.03598 and arXiv:1708.01588

Automated mathematics discovery run `fa_banach_001`
Agent `agent_lane_08` (GPT5.6)

July 23, 2026

Status. Candidate full solution, likely valid, pending expert review. The result gives a square, nonsingular matrix system for every positive support parameter and every classical compact symmetry type. It also proves the continuity and piecewise real-analyticity conjectured in arXiv:1708.01588, Conjecture 12.1.

Abstract

Freeman and Miller reduced optimal one-level test functions to a Fredholm equation and asked for an all-support algorithm that selects the correct coefficients from the delay-differential solution family. Freeman later obtained an all-support finite-dimensional family but left the coefficient optimization open. We give a direct finite algorithm. After translating the support interval to $[0, L]$, split it at the points of $(\mathbb{Z} \cup (L + \mathbb{Z})) \cap [0, L]$. Translation by one permutes the resulting cells in two chains. On each chain the differentiated Fredholm equation is a constant-coefficient ODE with a skew tridiagonal matrix. Matrix exponentials, continuity at cell boundaries, and one undifferentiated normalization equation form a square linear system. A strict operator-norm bound proves that this system is nonsingular for every $L > 0$. The system therefore computes the unique optimal factor and test function for all support restrictions. Analytic dependence of the finite system away from integer L, together with Hilbert–Schmidt resolvent continuity at integer L, proves the later regularity conjecture.

1 Source question and later sharpened problem

The source is Jesse Freeman and Steven J. Miller, *Determining Optimal Test Functions for Bounding the Average Rank in Families of L-Functions*, arXiv:1507.03598 [1]. In Section 6, page 16 of the arXiv PDF, they conjecture that their method provides solutions for all positive σ. They isolate two steps: solving the delay-differential system and selecting the coefficients that give the optimal factor g. They note that even an all-support algorithm for the first step would reduce the infinite-dimensional problem to finite dimensions.

6. Conclusion and Future Work

In [F] we include similar calculations for $3 < \sigma < 4$. We conjecture that these methods will provide solutions for all $\sigma \in \mathbb{R}^+$. In this pursuit, there are two important steps. The first is solving the system of delay differential equations. This gives a family of solutions. The second is taking the output of that system and finding the correct system of equations that give us the coefficients to pick our optimal g out of that family. Preliminary calculations suggest the second step will be more difficult than the first; however, even solving the first problem in general, or providing an algorithmic approach, is important progress as it reduces the problem to an optimization over a finite-dimensional space, as opposed to an infinite dimensional one.

Figure 1: The all-support conjecture and algorithmic problem in arXiv:1507.03598, Section 6, page 16. This is a direct crop from `source_paper.pdf`.

Freeman's later paper *Fredholm Theory and Optimal Test Functions for Detecting Central Point Vanishing Over Families of L-functions*, arXiv:1708.01588 [2], proves that the solution family is finite-dimensional for every σ. Corollary 9.9 computes its dimension. Section 12.1, page 62 (PDF page 63), says that the remaining finite optimization had only been solved for $1 < \sigma < 3/2$ and $3/2 < \sigma < 2$, asks for a general solution, and states:

"$\mathfrak{I}_G(\sigma)$ is continuous in σ and is real-analytic except at integers and half-integers."

12. Future Works

There are two clear directions for future work: remaining analysis of the one-level and analysis of optimal functions for the n-level, for $n > 1$. We discuss each in turn.

12.1. **The One-Level.** Due to restrictions on the number theory side of density computations, it is not as pressing to know all of the optimal test functions for all σ. However, our new method raises a few questions we feel should be addressed by future work.

First, in typical computations involving delay differential equations, one knows the function on an interval the size of the shift. For example, for the standard delay differential equation with a single delay given by

$$\frac{d}{dt}x(t) = f(x(t), x(t-\tau)) \tag{12.1}$$

we are given an initial condition such as $\phi : [-\tau, 0] \longrightarrow \mathbf{R}^n$. It can then be shown that the delay differential equation is equivalent to a homogenous initial value problem which can be solved via successive iteration [BC].

In this work we solve a system of location-specific delay-differential equations without any such initial condition or "history" of the solution. As this computation resembles solving a system of equations without the use of matrices, we are lead to ask if there is a general criterion for when such systems are solvable.

Second, after resolving this system of equations, we are left with a finite-dimensional optimization that leads to the optimal ϕ. We are able to solve two of these optimization problems ($1 < \sigma < 1.5$, and $1.5 < \sigma < 2$) using matrices. However, it is not obvious whether this method is necessary or sufficient. We seek a general solution to this finite-dimensional optimization problem.

Finally, there is much more to learn about the infimum function, $\mathfrak{I}_{\mathcal{G}}(\sigma)$. While we are able to show that the function is strictly decreasing, we suspect the following statement also holds.

Conjecture 12.1. The function $\mathfrak{I}_{\mathcal{G}}(\sigma)$ is continuous in σ and is real-analytic except at integers and half-integers.

62

Figure 2: The open coefficient problem and Conjecture 12.1 in arXiv:1708.01588, Section 12.1, printed page 62. This is a direct crop from `supporting_paper_1708.01588.pdf`.

2 Normalized Fredholm problem

Put $L = 2\sigma$ and translate $[-\sigma, \sigma]$ to $[0, L]$. Define the compact self-adjoint operator

$$(T_L f)(t) = \int_0^L \mathbf{1}_{\{|t-s|\le 1\}} f(s)\, ds \qquad (0 \le t \le L).$$

For $\beta \in \{1/2, -1/2\}$ set

$$A_{\beta,L} = I + \beta T_L.$$

The normalized optimality equations for SO(even) and Sp are respectively

$$A_{1/2,L} q_+ = \mathbf{1}, \qquad A_{-1/2,L} q_- = \mathbf{1}. \tag{2.1}$$

The other two symmetry types are rank-one consequences. If $P_L f = \langle f, \mathbf{1}\rangle\, \mathbf{1}$ and $Q_- = \int_0^L q_-(t)\, dt$, then

$$\begin{aligned} g_{\mathrm{SO(even)}} &= q_+, & g_{\mathrm{Sp}} &= q_-, \\ g_{\mathrm{SO(odd)}} &= \frac{q_-}{1+Q_-}, & g_{\mathrm{O}} &= \frac{\mathbf{1}}{1+L/2}. \end{aligned} \tag{2.2}$$

Indeed, the SO(odd) operator is $A_{-1/2,L} + P_L$, while the O operator is $I + \frac{1}{2} P_L$.

Lemma 1 (Strict norm bound)**.** *For every finite $L > 0$,*

$$\|T_L\|_{L^2(0,L)\to L^2(0,L)} < 2.$$

Consequently $A_{1/2,L}$ and $A_{-1/2,L}$ are positive and invertible.

Proof. The kernel of T_L belongs to $L^2([0,L]^2)$, so T_L is compact. Extend $f \in L^2(0,L)$ by zero to $\widetilde{f} \in L^2(\mathbb{R})$. The extension of $T_L f$ by zero is the compression to $[0,L]$ of $\mathbf{1}_{[-1,1]} * \widetilde{f}$. By Plancherel,

$$\left\| \mathbf{1}_{[-1,1]} * \widetilde{f} \right\|_2^2 = \int_{\mathbb{R}} |m(\xi)|^2 |\widehat{\widetilde{f}}(\xi)|^2\, d\xi,$$

where the convolution multiplier m satisfies $|m(\xi)| \le 2$, with equality only at the single frequency $\xi = 0$. Thus every nonzero f satisfies

$$\|T_L f\|_2 \le \left\| \mathbf{1}_{[-1,1]} * \widetilde{f} \right\|_2 < 2 \|f\|_2 .$$

If $\|T_L\| = 2$, compactness would give a unit vector at which the norm is attained, contradicting the strict inequality. Hence $\|T_L\| < 2$. Since T_L is self-adjoint, the spectra of $I \pm T_L/2$ lie in $(0,2)$. $\square$

3 The exact finite system

For $m \ge 1$, let $A_m(\beta)$ be the m-by-m skew tridiagonal matrix

$$(A_m)_{j,j+1} = -\beta, \qquad (A_m)_{j+1,j} = \beta \quad (0 \le j < m-1),$$

with all other entries zero. Put

$$E_m(t) = e^{tA_m(\beta)}, \qquad J_m(t) = \int_0^t e^{sA_m(\beta)}\, ds. \tag{3.1}$$

The formula for J_m is valid even when A_m is singular; it can be evaluated by one block-matrix exponential.

3.1 Noninteger length

Write $L = N + r$, where $N = \lfloor L \rfloor$ and $0 < r < 1$. Split $[0, L]$ into

$$R_k = [k, k + r] \quad (0 \le k \le N), \qquad S_k = [k + r, k + 1] \quad (0 \le k < N). \tag{3.2}$$

Let $u \in \mathbb{R}^{N+1}$ and $v \in \mathbb{R}^N$. For $N \ge 1$, determine them from the $2N + 1$ equations

$$(E_{N+1}(r)u)_k = v_k \qquad (0 \le k < N), \tag{3.3}$$

$$(E_N(1 - r)v)_k = u_{k+1} \qquad (0 \le k < N), \tag{3.4}$$

$$u_0 + \beta\big((J_{N+1}(r)u)_0 + (J_N(1 - r)v)_0\big) = 1. \tag{3.5}$$

Then reconstruct $q = q_{\beta,L}$ by

$$q(k + s) = (E_{N+1}(s)u)_k \qquad (0 \le k \le N,\ 0 \le s \le r), \tag{3.6}$$

$$q(k + r + s) = (E_N(s)v)_k \qquad (0 \le k < N,\ 0 \le s \le 1 - r). \tag{3.7}$$

When $0 < L < 1$, there is only one cell and the formula reduces to

$$q(t) = \frac{1}{1 + \beta L}. \tag{3.8}$$

3.2 Integer length

If $L = N \in \mathbb{N}$, determine $u \in \mathbb{R}^N$ from the N equations

$$(E_N(1)u)_k = u_{k+1} \qquad (0 \le k < N - 1), \tag{3.9}$$

$$u_0 + \beta(J_N(1)u)_0 = 1, \tag{3.10}$$

and set

$$q(k + s) = (E_N(s)u)_k \quad (0 \le k < N,\ 0 \le s \le 1). \tag{3.11}$$

Theorem 2 (All-support coefficient algorithm)**.** *For every $L > 0$ and $\beta \in \{1/2, -1/2\}$, the appropriate system* (3.3)–(3.5) *or* (3.9)–(3.10) *is square and nonsingular. Its reconstruction is the unique solution of*

$$(I + \beta T_L)q = \mathbf{1}. \tag{3.12}$$

Thus (2.2) *computes all four normalized optimal factors for every positive support parameter.*

Proof. We first prove exactness. Away from the breakpoints $(\mathbb{Z} \cup (L + \mathbb{Z})) \cap [0, L]$, differentiation of (3.12) gives

$$q'(t) = -\beta\, \mathbf{1}_{\{t+1<L\}} q(t + 1) + \beta\, \mathbf{1}_{\{t-1>0\}} q(t - 1). \tag{3.13}$$

Translation by one preserves each chain of cells in (3.2). If the restrictions of q to a chain are assembled into a vector $y(s)$ in left-to-right order, (3.13) is exactly

$$y'(s) = A_m(\beta)y(s). \tag{3.14}$$

Equations (3.3) and (3.4), or (3.9), are precisely the continuity conditions at consecutive cell boundaries.

For a reconstructed continuous function define

$$F(t) = q(t) + \beta \int_0^L \mathbf{1}_{\{|t-s|\le 1\}} q(s)\, ds.$$

Equation (3.14) gives $F'(t) = 0$ on every open cell. Since F is continuous, it is one constant on all of $[0, L]$. Equations (3.5) and (3.10) are exactly $F(0) = 1$, so the reconstruction solves (3.12).

Conversely, an L^2 solution of (3.12) is continuous because the moving window integral is continuous. On each open cell it is differentiable and satisfies (3.13), hence (3.14), and therefore yields endpoint data satisfying the displayed finite system.

It remains to prove nonsingularity. A vector in the kernel of the homogeneous finite system would reconstruct a solution of $(I + \beta T_L)q = 0$. Lemma 1 forces $q = 0$, and the endpoint vector is then zero. The square coefficient matrix has trivial kernel and is nonsingular. Existence and uniqueness follow. $\square$

Remark 3 (Dimension check). Off integer L, the system has $2\lfloor L \rfloor + 1$ unknowns. At integer L it has L unknowns. Since $L = 2\sigma$, these become

support range	dimension
$k < \sigma < k + \frac{1}{2}$	$4k + 1$
$k + \frac{1}{2} < \sigma < k + 1$	$4k + 3$
$\sigma = k$	$2k$
$\sigma = k + \frac{1}{2}$	$2k + 1,$

which agrees exactly with Freeman's Corollary 9.9 [2]. The new equations select the unique coefficient vector in that family.

4 Proof intuition

The source difficulty is that differentiating the Fredholm equation loses one scalar constant and appears to produce a delay equation without an initial history. The breakpoints remove both obstacles. They are exactly the places at which either $t + 1$ or $t - 1$ enters or leaves the support. Between them, translation by one never changes combinatorial type, so all translated pieces of q evolve together by one constant skew tridiagonal matrix.

Matrix exponentials therefore solve the delay system without prescribing an arbitrary history. Continuity glues the cell solutions, while a single undifferentiated equation at $t = 0$ restores the scalar constant lost under differentiation. Finally, Fredholm invertibility turns the resulting square system from a candidate construction into a guaranteed algorithm: a singular coefficient matrix would manufacture a forbidden homogeneous Fredholm solution.

5 Optimality and the test function

The source papers identify the admissible Fourier transforms as

$$\widehat{\phi} = g * \breve{g}, \qquad \breve{g}(x) = \overline{g(-x)}, \qquad \operatorname{supp} g \subset [-\sigma, \sigma]. \tag{5.1}$$

For completeness, optimality of the factor produced above follows directly from positivity. Let B_G be the corresponding positive operator:

$$\begin{aligned} B_{\mathrm{SO(even)}} &= I + \tfrac{1}{2}T_L, & B_{\mathrm{Sp}} &= I - \tfrac{1}{2}T_L, \\ B_{\mathrm{SO(odd)}} &= I - \tfrac{1}{2}T_L + P_L, & B_{\mathrm{O}} &= I + \tfrac{1}{2}P_L. \end{aligned}$$

If $g_G = B_G^{-1}\mathbf{1}$, then for every nonzero admissible factor h, Cauchy–Schwarz in the B_G inner product gives

$$|\langle h, \mathbf{1}\rangle|^2 = \left|\left\langle B_G^{1/2}h, B_G^{-1/2}\mathbf{1}\right\rangle\right|^2 \le \langle B_G h, h\rangle\ \langle g_G, \mathbf{1}\rangle\,. \tag{5.2}$$

Hence

$$\frac{\langle B_G h, h\rangle}{|\langle h, \mathbf{1}\rangle|^2} \geq \frac{1}{\int_0^L g_G(t)\,dt}, \tag{5.3}$$

with equality exactly for scalar multiples of g_G. The normalization $B_G g_G = \mathbf{1}$ fixes the scalar. Translating g_G back to $[-\sigma, \sigma]$ and applying (5.1) yields the optimal test function

$$\phi_G = \left|\mathcal{F}^{-1} g_G\right|^2. \tag{5.4}$$

The optimal infimum is

$$\mathfrak{I}_G(\sigma) = \frac{1}{\int_0^L g_G(t)\,dt}. \tag{5.5}$$

In particular, if $Q_\pm = \int_0^L q_\pm$, then

$$\begin{aligned} \mathfrak{I}_{\mathrm{SO(even)}}(\sigma) &= Q_+^{-1}, & \mathfrak{I}_{\mathrm{Sp}}(\sigma) &= Q_-^{-1}, \\ \mathfrak{I}_{\mathrm{SO(odd)}}(\sigma) &= 1 + Q_-^{-1}, & \mathfrak{I}_{\mathrm{O}}(\sigma) &= \frac{1}{L} + \frac{1}{2}. \end{aligned} \tag{5.6}$$

6 Full regularity classification

Theorem 4 (Conjecture 12.1)**.** *For each $G \in \{\mathrm{SO(even)}, \mathrm{Sp}, \mathrm{SO(odd)}, \mathrm{O}\}$, the function $\mathfrak{I}_G : (0, \infty) \to \mathbb{R}$ is continuous. It is real-analytic on every component of*

$$(0, \infty) \setminus \tfrac{1}{2}\mathbb{Z}.$$

Thus Freeman's Conjecture 12.1 holds. For $G = \mathrm{O}$, the function is in fact real-analytic on all of $(0, \infty)$.

Proof. Fix an integer $N \geq 0$. For $L = N + r$ with $0 < r < 1$, every entry of the system matrix in (3.3)–(3.5) is real-analytic in r: its entries are formed from e^{rA}, $e^{(1-r)A}$, and their integrals. The determinant never vanishes by Theorem 2. Cramer's rule, or analytic matrix inversion, shows that u, v, and

$$Q_{\beta,L} = \mathbf{1}^\mathsf{T} J_{N+1}(r) u + \mathbf{1}^\mathsf{T} J_N(1-r) v \tag{6.1}$$

are real-analytic in r (with the evident one-cell formula when $N = 0$). The formulas (2.2) and (5.6) preserve real-analyticity. Positivity guarantees that every denominator is nonzero.

It remains to cross integer values of L. Work on the fixed space $L^2(\mathbb{R})$ and extend the interval operator by zero. Its kernel is

$$k_L(x, y) = \mathbf{1}_{[0,L]}(x)\mathbf{1}_{[0,L]}(y)\mathbf{1}_{\{|x-y|\leq 1\}}. \tag{6.2}$$

As $L' \to L$, $k_{L'} \to k_L$ in $L^2(\mathbb{R}^2)$, so the corresponding operators converge in Hilbert–Schmidt, hence operator, norm. Also $\mathbf{1}_{[0,L']} \to \mathbf{1}_{[0,L]}$ in $L^2(\mathbb{R})$. Since $I \pm T_L/2$ is invertible, the resolvent identity gives local operator-norm continuity of its inverse. Therefore the zero extensions

$$q_{\pm,L} = (I \pm T_L/2)^{-1}\mathbf{1}_{[0,L]}$$

vary continuously in $L^2(\mathbb{R})$. Their integrals are the inner products $\left\langle q_{\pm,L}, \mathbf{1}_{[0,L]}\right\rangle$ and are continuous. Equations (2.2) and (5.6) then prove continuity for every symmetry type. Finally, $L = 2\sigma$, so integer breakpoints in L are precisely the integers and half-integers in σ. □

7 Computational verification

The included script `code/verify_algorithm.py` independently builds the breakpoint orbits, constructs the square system through matrix exponentials, and evaluates the Fredholm residual on a 401-point grid. It tests both signs of β and the SO(odd) rank-one rescaling.

Tested L values	Main maximum residual	Additional check
$0.4, 1, 1.4, 2, 2.4, 3, 3.7, 5.25, 8.1$	$< 4.3 \cdot 10^{-14}$	both $\beta = \pm 1/2$
$1 \pm 10^{-6}, 2 \pm 10^{-6}, 3 \pm 10^{-6}$	$< 2 \cdot 10^{-8}$	one-sided integral continuity

Run from the repository root with

```
conda run --no-capture-output -n sandbox python \
  runs/fa_banach_001/solutions/full/\
1507.03598_all_support_optimal_test_function_algorithm/code/verify_algorithm.py
```

The numerical calculation is not used as proof; it pressure-tests signs, cell indexing, continuity equations, normalization, and the rank-one formula.

8 Novelty check, scope, and review priorities

Bounded search. On July 23, 2026, the local registry, solution, attempt, and proof-gap indexes were searched for arXiv ids 1507.03598 and 1708.01588 and for the core phrases "optimal test function", "finite-dimensional optimization", and "real-analytic except at integers and half-integers". Bounded arXiv searches used the two arXiv ids, the authors Freeman and Miller, "matrix exponential", "all support", "coefficient", and "Conjecture 12.1". The search found Freeman's 2017 all-support finite-dimensional reduction [2] and the later restricted two-level extension arXiv:2011.10140 [3]. No paper was found that gives the square all-support coefficient system above or states a proof of Conjecture 12.1. This supports, but cannot guarantee, novelty.

Scope. The theorem solves the one-level compact-group optimization problem with the indicator kernels used in the source papers. It does not verify a number-theoretic density theorem at new support, incorporate lower-order terms, or solve unrestricted higher-level density optimization.

Human-review priorities. The most valuable checks are:

1. verify the translation from the source interval and kernels to (2.1)–(2.2);
2. check the two cell-chain endpoint conventions in (3.3)–(3.7);
3. confirm that the source admissibility/factorization theorem permits the variational conclusion (5.4) at every positive support;
4. search citations to arXiv:1708.01588 for a later unpublished or differently phrased solution.

References

[1] Jesse Freeman and Steven J. Miller, *Determining Optimal Test Functions for Bounding the Average Rank in Families of L-Functions*, arXiv:1507.03598, 2015. https://arxiv.org/abs/1507.03598.

[2] Jesse Freeman, *Fredholm Theory and Optimal Test Functions for Detecting Central Point Vanishing Over Families of L-functions*, arXiv:1708.01588, 2017. https://arxiv.org/abs/1708.01588.

[3] Elzbieta Boldyriew, Fangu Chen, Charles Devlin VI, Steven J. Miller, and Jason Zhao, *Determining Optimal Test Functions for* 2*-Level Densities*, arXiv:2011.10140, 2020. https://arxiv.org/abs/2011.10140.